Xiangling Xu 许湘灵

PhD student, PhIQuS, Inria Saclay

My Website
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Google Scholar
ORCID



Research Interests

I am interested in the capabilities and limits of quantum physics. To explore these questions, I study correlations in quantum networks, certifiable properties of quantum many-body systems, and cryptographic approaches to testing quantum devices. My main mathematical tools are the theory of operator algebras and noncommutative polynomial optimization.

Education

Ph.D. in Computer Science, Institut Polytechnique de Paris 2023 – Present
Project: *Quantum correlations in causal structures: characterization and application.*
Expected completion: 2027-05; supervisor: Marc-Olivier Renou.

M.Sc. Mathematics, ETH Zürich 2020 – 2023
Thesis: *Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective.*
Supervisor: Renato Renner (Institute for Theoretical Physics); GPA: 5.69/6.00.

B.Sc. Mathematics and Physics, University of Toronto 2016 – 2020
Mathematics and Physics Specialist Program; GPA: 3.95/4.00.
Selected research projects: large-scale cosmological structures, light trapping in plasmonic nanogratings.

Publications and Preprints

1. **An operational characterization of finite-dimensional quantum theory**.
L. J. Dmello, X. Xu, M.-O. Renou, and D. Gross.
2026-10. arXiv: 2610.03694 [quant-ph].

2. **Quantum inflation and source-transfer hierarchies are complete for tree networks**.
X. Xu, I. Klep, and M.-O. Renou.
2026-09. arXiv: 2609.19310 [quant-ph].

3. **Fermions are fundamentally more nonlocal than Bosons**.
F. M. Kalarde, S. Boreiri, X. Xu, L. Tendick, S. Beigi, P. Perinotti, T. Guaita, and M.-O. Renou.
2026-06. arXiv: 2606.12363 [quant-ph].

4. **The bulk spectral gap is certifiable from above but uncomputable from below**.
X. Xu, M. Schötz, J. Wang, V. Magron, I. Klep, O. Fawzi, and M.-O. Renou.
2026-06. arXiv: 2606.03836 [quant-ph].

5. **Comment on "Quantum theory based on real numbers cannot be experimentally falsified": on the compatibility of physical principles with information theory for fermions**.
F. M. Kalarde, X. Xu, and M.-O. Renou.
2026-04. arXiv: 2604.07425 [quant-ph].

6 **Two convergent NPA-like hierarchies for the quantum bilocal scenario**.
M.-O. Renou, X. Xu, and L. T. Ligthart.
Journal of Mathematical Physics, vol. 67, no. 1, Art. no. 012203, 2026-01. DOI: 10.1063/5.0211008

7 **Quantitative quantum soundness for all multipartite compiled nonlocal games**.
M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, and X. Xu.
2025-09. arXiv: 2509.25145 [quant-ph]. QIP 2026 (contributed talk); alphabetical author order.

8 **Quantitative quantum soundness for bipartite compiled Bell games via the sequential NPA hierarchy**.
I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, X. Xu, and Y. Zhao.
2025-07. arXiv: 2507.17006 [quant-ph]. QIP 2026 (contributed talk); alphabetical author order.

9 **Quantitative Tsirelson's theorems via approximate Schur's lemma and probabilistic Stampfli's theorems**.
X. Xu, M.-O. Renou, and I. Klep.
2025-05. arXiv: 2505.22309 [quant-ph].

10 **On the kinematic morphology around haloes**.
X. Luo, X. Xu, and X. Wang.
Monthly Notices of the Royal Astronomical Society, vol. 518, no. 4, pp. 6059–6064, 2023. DOI: 10.1093/mnras/stac3500

11 **Quantum nonlocality in bilocal networks: an operator algebraic perspective**.
M.S. thesis, ETH Zurich, 2023. DOI: 10.3929/ethz-b-000613670

Selected Talks

My work has been presented at leading international conferences in quantum information (QIP), quantum cryptography (QCrypt), optimization (SIAM), and operator theory (IWOTA), and in selected contributed talks at GDR TeQ, a major national meeting of France's CNRS Quantum Technologies network.

The bulk spectral gap is RE-complete: certifiable from above, uncomputable from below

2026-11 Journées Informatique Quantique, Toulouse, France (contributed).
Gif-sur-Yvette, France (contributed).

Quantitative quantum soundness for compiled nonlocal games and the sequential NPA hierarchy

2026-08 QCrypt, Ottawa, Canada (contributed).
2026-06 SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom (invited).
2026-01 QIP 2026, Riga, Latvia (two contributed talks: bipartite and multipartite games).
2025-11 3rd Colloquium GDR TeQ, Grenoble, France (contributed).
2025-10 YQIS25, Barcelona, Spain (contributed).
2025-07 IWOTA, Enschede, Netherlands (invited).
2025-06 IQC-PCQT-Quantum Saclay Workshop, Paris, France (invited).

Characterizing quantum bilocal networks with generalized NPA hierarchies

2024-09 CausalWorlds, Waterloo, Canada (contributed).
2023-11 Defi EQIP Workshop, Lyon, France (contributed).
2023-10 Saclay Quantum Seminar, Palaiseau, France (invited).

Selected Talks (continued)

Other research seminars

2026-11 Institute for Theoretical Physics, University of Cologne.
2026-09 Quriosity, Inria / Télécom Paris / IP Paris.
2026-08 Quantum Science Center of Guangdong-Hong Kong-Macao Greater Bay Area (Ranyiliu Chen).
2026-07 Nanjing University (Penghui Yao).
2026-02 Institut de physique théorique, CEA (Jean-Daniel Bancal).
2025-12 Quantum Meets, online (invited).
AMSS, Chinese Academy of Sciences (Jie Wang).
2025-11 EPFL (Thomas Vidick).
2025-10 Freie Universität Berlin (Jens Eisert).
2025-02 QAT team, ENS Paris.
2024-07 University of Waterloo (William Slofstra).
2023-12 University of Cologne (David Gross).

Posters

2026 CQCC & GQSF, Shenzhen, China.
2024 2nd Colloquium GDR TeQ, Paris, France; YQIS24, Paris, France.

Research Visits

2026-10 – 2026-11 David Gross — University of Cologne.
2026-07 Penghui Yao — Nanjing University.
2026-05, 2025-12 Jie Wang — AMSS, Chinese Academy of Sciences.
2025-10 – 2025-11 Thomas Vidick — EPFL.
2024-09 Igor Klep — University of Ljubljana.
2024-07 William Slofstra — Institute for Quantum Computing, University of Waterloo.
2023-12 David Gross — University of Cologne.

Academic Service

Organization

2024-02 – Present **Co-organizer, COMPUTE (QuantERA)**
Five-partner European collaboration on quantum networks and noncommutative polynomial optimization (Inria Saclay, LAAS-CNRS Toulouse, TU Darmstadt, University of Ljubljana, ICFO; EUR 0.99 million total project funding).
- Contributed to the successful QuantERA proposal and coordinated consortium meetings and project dissemination.

2025-03 – 2026-03 **Co-organizer, Gaspard Monge Visiting Professorship and seminar series**
Visiting professor Igor Klep; seminars on polynomial optimization and quantum theory, École Polytechnique and Inria Saclay.
- Helped draft the funding application and organized invited talks by Jean-Bernard Lasserre, Monique Laurent, Antonio Acín, Hamza Fawzi, William Slofstra, and Andreas Bluhm.

Academic Service (continued)

2025-03 **Co-organizer, 828th WE-Heraeus Seminar**
Operator Theory and Polynomial Optimization in Quantum Information Theory, Physikzentrum Bad Honnef.

2023-09 **Co-organizer, Saclay Quantum Workshop**
One-day workshop on Navascués–Pironio–Acín (NPA) hierarchies.

Peer Review

Referee: *Quantum* (journal), *Operator Theory* (Springer reference work; chapter), QIP, CausalWorlds.

Selected Awards and Funding

2026 **Erasmus+ Mobility Funding** EUR 1,154
Research visit: University of Cologne.

2024 **Erasmus+ Mobility Funding** EUR 1,274
Research visit: University of Ljubljana.

2018 **Class of 3To and Associates Scholarship in Mathematics and Physics** CAD 2,050
University of Toronto · Selective merit scholarship recognizing outstanding academic achievement in second- or third-year Mathematics and Physics.

International Experience (IE) Award CAD 2,500
University of Toronto · Summer research exchange at the University of Liverpool.

2017 **Summer Undergraduate Research Program (SURP)** CAD 9,000
Canadian Institute for Theoretical Astrophysics (CITA) · Selected for a 16-week paid research program recruiting undergraduates internationally, as the only first-year student in the 2017 cohort. The project led to a publication.

2016 **President's Scholar of Excellence** CAD 10,000
University of Toronto · One of approximately 50 university-wide recipients (over 43,000 first-entry undergraduate admissions in 2016), with additional benefits such as faculty mentorship.

Teaching Experience

2024, 2025 **Guest lecturer** (one lecture per year), *Recent Trends in Parallel, Distributed, and Quantum Computing*, Quantum and Distributed Computer Science (QDCS) Master's program, Université Paris-Saclay.

2022 – 2023 **Teaching assistant**, *Linear Algebra I & II*, ETH Zürich.

2021 – 2022 **Teaching assistant**, *Analysis I & II*, ETH Zürich.

Skills

Computing Python, MATLAB, C, Mathematica; ncpol2sdpa, YALMIP.

Languages Chinese (native), English (fluent), German (beginner), French (beginner).